

Physical Examination of the Abdomen

Preparation Questions

1. What are the regions and quadrants of the abdomen? What organs are located in each abdominal quadrant?
2. What is the sequence of examination of the abdomen?

Before the start of this examination, you must:

- Wash hands or use hand sanitizer - Ensure that sanitizer dries completely
- Greet and introduce yourself to your patient
- Identify your patient (enquire about patient's name)
- Explain to your patient what you are going to do before and during the examination
- Obtain permission from your patient before beginning the physical exam
- Properly position the patient and expose the area to be examined

Remember that, at all times, you should demonstrate professionalism and show the same respect to your patients and colleagues that you would to your future patients.

Patient position and exposure

Standing at the patient's right side at the bed or examination table, with the patient lying supine, with their back flat on the bed. Ensure that they are as comfortably as possible. You may ask the patient to flex their knees such that the feet are flat on the bed or table (this allows for a flat lumbar area and relaxed abdominal muscles). Expose the abdomen by lowering the gown to the nipple line and lowering the drape to the suprapubic level.

Inspection

Observe the patient from the foot of the bed then stand on the right side of the patient inspecting the abdomen for:

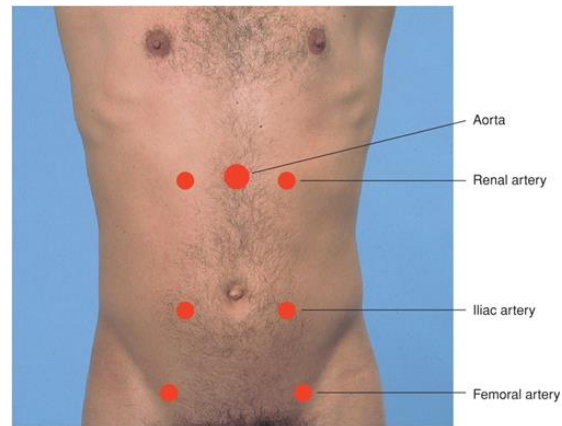
- **Symmetry and contour:** Is the abdomen flat, scaphoid (concave), rounded, or protuberant? Look for bulges or localized swelling. Look for visible organs (such as enlarged liver and or spleen) or visible masses.
- **Skin:** scars (surgical or other); striae, rashes, lesions, dilated

- superficial veins or ecchymosis.
- **Umbilicus:** contour and location and any inflammation or bulges.
- **Abnormal movements:** aortic pulsations and even intestinal peristaltic movements can be normally visible in children and very thin adults, but are otherwise worrisome for aortic aneurysm or intestinal obstruction
- **Hernias:** ask the patient to turn his/her head to the side and cough. A bulge or swelling may indicate a hernia

Before touching the patient's abdomen, you must ask about any areas of pain or tenderness and begin touching the abdomen in an area away from the pain.

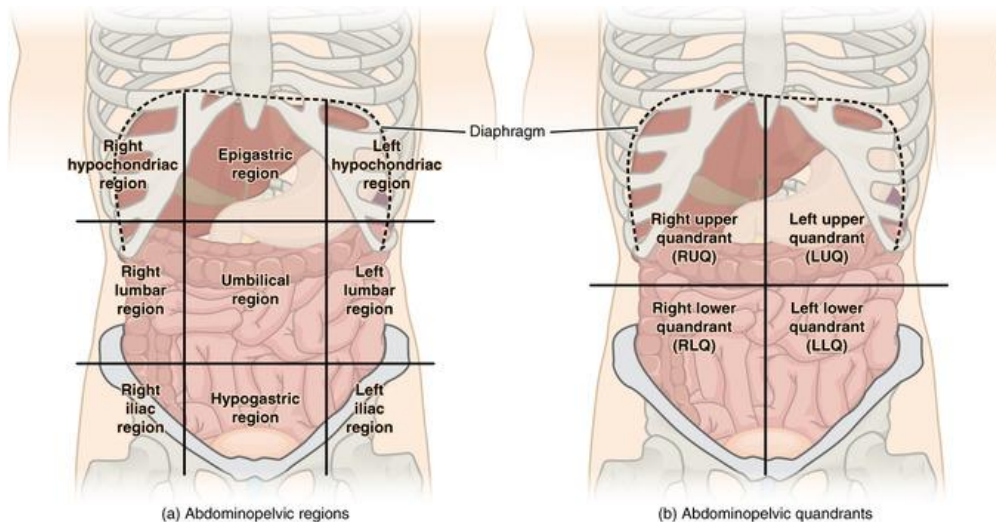
Auscultation (auscultation precedes percussion/palpation in abdominal examination)

- **Auscultate for bowel sounds:** Using the diaphragm of your stethoscope, note frequency and character. Auscultating in the RLQ is sufficient. Normal bowel sounds of peristalsis occur approximately every 5 to 10 seconds and have a low-pitched gurgle-like sound
 - If bowel sounds are present and normal, there is no need to listen to other quadrants
 - If diminished or absent, listen in each quadrant.
- **Listen for bruits:** Using the diaphragm of your stethoscope, listen at the following areas on your patient
 - Aorta:** 5cm (2 inches) above the umbilicus
 - Right Renal artery:** 5cm (2 inches) above the umbilicus then 2.5cm (1 inch) on the patient's right side.
 - Left Renal arteries:** 5cm (2 inches) above the umbilicus then 2.5cm (1 inch) to the patient's left side
 - Right Iliac artery:** ~2 inches below the umbilicus on a line to the right femoral pulse
 - Left Iliac artery:** ~2 inches below the umbilicus on a line to the left femoral pulse
 - **Femoral arteries – NOT DONE IN LAB**



General Percussion

- **Demonstrate Proper Percussion technique:** To perform percussion, hyperextend the middle finger of your nondominant hand with the distal interphalangeal joint [DIP] pressed on the skin surface to be explored. This is known as the pleximeter finger. Avoid contact with other parts of the hand. Flex the middle finger of your other hand (the dominant hand) in a striking position; this is the plexor finger. Using a quick, sharp motion, strike the pleximeter finger with the tip of the plexor finger. The motion is a brisk but relaxed flexion of the wrist. Strike twice in each location, then move to the next location.
- Percuss generally in all four quadrants / **nine regions - MUST** to determine the distribution of tympany and dullness. Note any tenderness.



General Palpation

Superficial Palpation

- Perform light palpation in all four quadrants/nine regions. This should be

superficial, and any area of reported pain should be assessed last. Place one (dominant) hand flat on the abdomen in the horizontal plane with fingers together. Gently press into the abdomen. using movements at the IP joints. Then slide the hand to the next location, repeating the movement.

- Identify any area of tenderness, masses, guarding or rigidity.



Deep Palpation

- Perform deep palpation in **all** four quadrants / **nine regions** – **MUST** . Place your dominant hand flat on the abdomen and place your nondominant hand on top then press down deeply (1.5-2 inches). It is not sufficient to palpate one area in each quadrant
- Identify and describe palpable masses regarding their: Location, Size, Shape, Consistency (cystic, soft, firm, solid), Tenderness, Pulsation and Mobility.



Special Test:

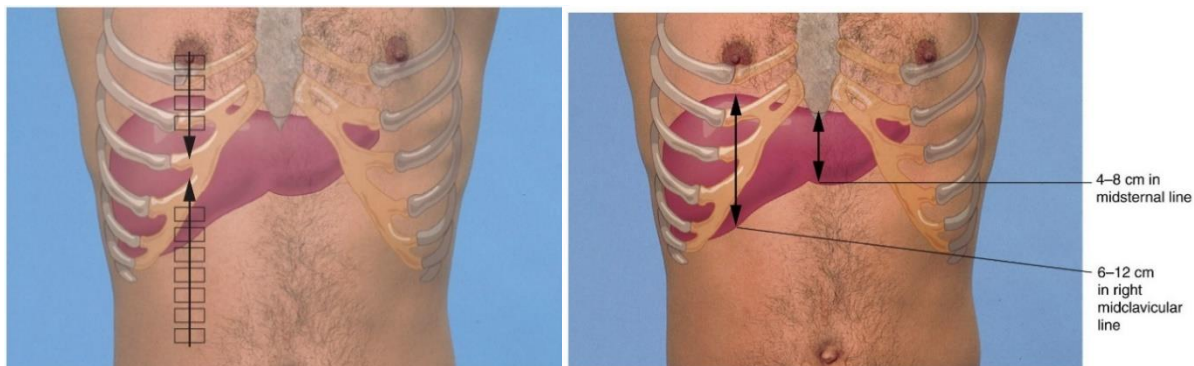
Rebound Tenderness

Press down slowly in the area of tenderness, then withdraw your hand quickly. Ask patient if pain was worst on pressing or release. The maneuver is positive if withdrawal produces more pain.

Examination of the Liver

Percussion

- **Measure the Liver span.**
 - Percuss from the RLQ in the mid- clavicular line up towards the liver. Note the change from tympany to dullness and mark this point – Lower liver border.
 - Next, percuss at the nipple line **or below the breast in the adult female** in the midclavicular line downwards and note the change from lung resonance to dullness. Mark this point – Upper liver border.
 - Measure the difference between the two points



Palpation

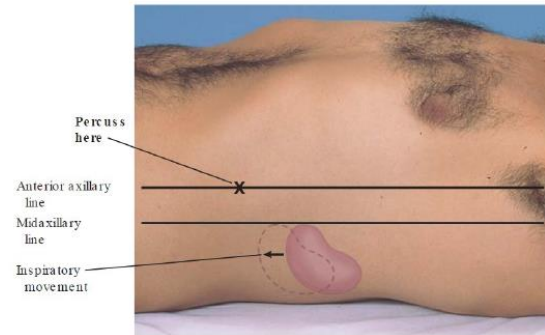
- Place your left hand posteriorly between the right twelfth rib and the iliac crest, gently pressing anteriorly
- Place the fingertips of the right hand in the RLQ parallel and lateral to the rectus muscle in the midclavicular line.
- Ask patient to take a deep breath during the exam. You may feel the liver edge descend during inspiration.
- Move upwards and press inward depressing the abdomen during expiration. Ask the patient to take a deep breath and feel for the liver edge.



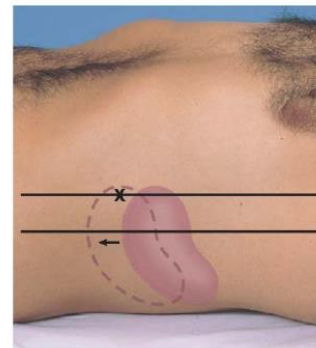
Examination of the Spleen

Percussion

- Percuss at the anterior axillary line at the lowest interspace. Note any tympany. Then ask patient to take a deep breath while continuing to percuss and note any changes. When the spleen size is normal, the percussion note is tympanic. If there is splenomegaly, the note is dull.



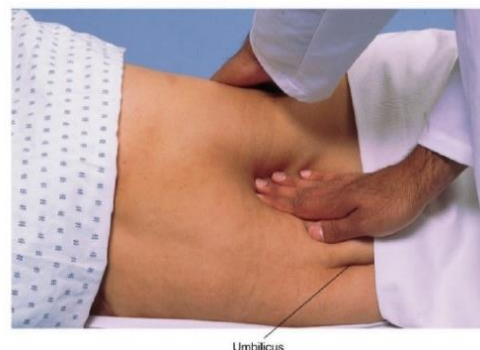
Negative splenic percussion sign



Positive splenic percussion sign

Palpation

- Reach over, place the left hand behind the patient's left lower ninth, tenth, and eleventh ribs and push upward. Using right hand, start palpation in the RLQ and palpate in an oblique angle towards the spleen. Ask patient to take a deep breath and try to feel for the spleen edge on inspiration.
- If the spleen is still not palpable, ask the patient to turn on the right side and repeat the palpation technique



Examination of the Kidneys

Palpation (ballottement)

- Place your right hand behind the patient, just below and parallel to the 12th rib and push upwards to displace the left kidney anteriorly. With your left hand lateral and parallel to the rectus muscle, ask the patient to breathe in deeply and press firmly downward in the LUQ along the midclavicular line trying to capture the kidney.
- Place your left hand behind the patient, just below and parallel to the 12th rib and push upwards to displace the right kidney anteriorly. With your right hand lateral and parallel to the rectus muscle, ask the patient to breathe in deeply and press firmly downward in the RUQ along the midclavicular line trying to capture the kidney.



Checklist:

Complete the table while assessing your peer(s) during SG session and later when engaging in independent review.

Categories	Task	Performed/Not Performed
Professionalism	Sanitize your hands	
Professionalism	Greet patient	
Professionalism	Introduce yourself by stating your name and that you are a Term 2 medical student	
Professionalism	Ask the patient's name and how they would like to be addressed	
Professionalism	Ask if the patient is having pain anywhere in the abdomen. If yes, have them point to the area of pain	
Professionalism	Explain procedure: E.g. Tell the patient 'I am going to listen to your abdomen with my stethoscope. E.g. I am going to be touching your abdomen with my hands. E.g. I am going to be tapping on your abdomen. E.g. I am going to feel for your liver by tapping on your abdomen. E.g. I am going to feel for your liver by touching your abdomen. E.g. I am going to feel for your spleen by tapping your abdomen. E.g. I am going to feel for your spleen by touching your abdomen. E.g. I am going to feel for your kidneys by touching your abdomen.	
Professionalism	Obtain consent: (ask 'Is that ok?')	
Professionalism	Properly position the patient and expose the area to be examined	

Professionalism	Stand at the right side of the patient	
Inspection	Observe the patient from the foot of the bed then stand on the right side of the patient looking for symmetry and contour: flat, protuberant, rounded, scaphoid (concave), flanks bulge or localized swelling; skin: scars, striae, rashes, lesions, dilated veins or ecchymosis; umbilicus: contour and location and any inflammation or bulges; abnormal movements: peristalsis or aortic pulsations.	
Auscultation for bowel sounds	Place the diaphragm of the stethoscope on the abdomen in the right lower quadrant and listening to note the frequency and character of the bowel sounds. If bowel sounds are present and normal, there is no need to listen to other quadrants. If bowel sounds are diminished or absent, listen in each quadrant. If pain in the right lower quadrant, auscultate first in any of the other quadrants.	
Auscultation for bruits	Place the diaphragm of your stethoscope on the abdomen in the following areas: Aorta: 5cm (2 inches) above the umbilicus Right Renal artery: 5cm (2 inches) above the umbilicus then 2.5cm (1 inch) to the patient's right side Left Renal arteries: 5cm (2 inches) above the umbilicus then 2.5cm (1 inch) to the patient's left side Right Iliac artery: ~2 inches below the umbilicus on a line to the right femoral pulse Left Iliac artery: ~2 inches below the umbilicus on a line to the left femoral pulse	
General Percussion	Hyperextend the middle finger of your nondominant hand with the distal interphalangeal joint [DIP] pressed on the surface of the abdomen (this is the pleximeter finger). Avoid contact with other parts of the hand. Flex the middle finger of your other hand (the dominant hand) in a striking position (this is the plexor finger). Using a quick, sharp motion of the wrist, strike the pleximeter finger with the tip of the plexor finger. Strike twice in each location, then move to the next location. Percuss in all 4 quadrants listening for areas of tympany and dullness. Start percussing away from any areas of pain.	
Superficial Palpation	Place one hand flat on the abdomen in the horizontal plane with fingers together. Gently press into the abdomen. using movements at the IP joints. Then slide the hand to the next location, repeating the movement. Identify any area of tenderness, masses, guarding or rigidity Start palpating away from any areas of pain. While palpating, observe patient's face for any signs of pain.	

Deep Palpation	Place your dominant hand flat on the abdomen and place your nondominant hand on top then press down deeply (1.5-2 inches). Palpate in all 4 quadrants. Palpate all 4 quadrants for masses regarding their location, size, shape, consistency (cystic, soft, firm, solid), tenderness, pulsation and mobility. Start palpating away from any areas of pain. While palpating, observe patient's face for any signs of pain.	
Test for Rebound Tenderness	Press down slowly in the area of tenderness, then withdraw your hand <u>quickly</u> . Ask patient if pain was worst on pressing or release. While palpating, observe patient's face for any signs of pain.	
Percussion of the Liver	Percuss from the right lower quadrant of the abdomen in the mid-clavicular line up towards the liver. Note the change from tympany to dullness and mark this point as the lower liver border.	
Percussion of the Liver	Percusses at the nipple line or below the breast in the adult female in the midclavicular line downwards and note the change from lung resonance to dullness.	
Palpation of the Liver	Places your left hand posteriorly between the right twelfth rib and the iliac crest, gently pressing anteriorly. Place the fingertips of the right hand in the right lower quadrant of the abdomen parallel and lateral to the rectus muscle in the midclavicular line.	
Palpation of the Liver	Ask the patient to take a deep breath then presses fingers in a dipping motion feeling for the liver. Move upwards and press inward depressing the abdomen during expiration. As the patient to take a deep breath and feel for the liver edge.	
Percussion of the Spleen	Percuss at the anterior axillary line at the lowest interspace. Note any tympany.	
Percussion of the Spleen	Ask the patient to take a deep breath while continuing to percuss and note any changes	
Palpation of the Spleen	Reach over and places the left hand behind the patient's left lower ninth, tenth, and eleventh ribs and push upwards.	
Palpation of the Spleen	Using the right hand, start palpation (hand on abdomen using movements of the IP joints) in the right lower quadrant of the abdomen and palpate in an oblique angle towards the spleen. Ask patient to take a deep breath and try to feel for the spleen edge on inspiration.	

Palpation of Left Kidney	Place your right hand behind the patient, just below and parallel to the 12 th rib and push upwards to displace the left kidney anteriorly. With your left hand lateral and parallel to the rectus muscle, ask the patient to breathe in deeply and press firmly downward in the LUQ along the midclavicular line trying to capture the kidney.	
Palpation of Right Kidney	Place your left hand behind the patient, just below and parallel to the 12 th rib and push upwards to displace the right kidney anteriorly. With your right hand lateral and parallel to the rectus muscle, ask the patient to breathe in deeply and press firmly downward in the RUQ along the midclavicular line trying to capture the kidney.	